

Question Paper

Exam Date & Time: 05-Jan-2026 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - JANUARY 2026

Operating Systems [ECM_3123]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

1) (5)

A)

Given the following set of processes, their arrival times, burst times, and priorities:

Processes	Arrival Time	Burst Time	Priority
P1	0	5	-1
P2	1	3	-2
P3	2	8	3
P4	3	6	1
P5	4	5	2
P6	5	2	-2

a) Draw the Gantt chart for preemptive priority scheduling (if two processes have the same priority, schedule the low process id first). (Lower number indicates Lower priority)

b) Draw the Gantt chart for MRTF (Minimum Remaining Time First).

Course Outcome: "CO2" / Cognitive Level: "Application"

B) Consider the following code where two processes, proc1 and proc2, concurrently update a shared integer variable counter initialized to 0: (5)

```
int counter = 0;
proc1 {
  int X = counter;
  counter = X - 1;
  X = counter;
  counter = X - 1;
}
```

```

proc2 {
int Y = counter;
counter = Y - 1;
}

```

1. What is the expected final value of counter if the operations of proc1 and proc2 are executed sequentially, without interruption or overlapping?
2. List all possible final values of counter when proc1 and proc2 execute concurrently without any synchronization.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 2) **Given the following data for a system with 6 processes (P0 to P5) and 4 resource types (A, B, C, D):** (5)

A)

Process	Allocation	Maximum
	A B C D	A B C D
P0	1 2 2 1	3 3 2 2
P1	2 0 1 1	2 1 2 2
P2	3 1 1 0	5 2 3 1
P3	2 1 2 1	3 3 4 2
P4	0 0 1 2	2 2 2 2
P5	1 1 0 0	3 3 3 2

Available resources:

A = 3, B = 2, C = 1, D = 2

- a) Construct the *Need* matrix for the given data.
- b) Using the Banker's Algorithm, determine whether the system is currently in a safe state. If so, provide safe sequences.
- c) If process P2 requests 1 instance of resource type A and 1 instance of D, can this request be granted immediately? Justify using Banker's Algorithm.

Course Outcome: "CO4" / Cognitive Level: "Remember"

- B) A system contains 5 processes (P1, P2, P3, P4, P5) and 3 resources (R1, R2, R3) with the following number of instances: (5)

- R1: 2 instances
- R2: 3 instances
- R3: 2 instances

At a particular moment, the system's resource allocation status is:

- P1 holds 1 instance of R1 and requests 1 instance of R2.
- P2 holds 1 instance of R2, requests 1 instance of R3.
- P3 holds 1 instance of R3, requests 1 instance of R1.
- P4 holds 1 instance of R2 and is not requesting any other resource.
- P5 is requesting 1 instance of R2 and 1 instance of R3 (but holds none currently).

- a) Draw the resource allocation graph, clearly indicating all allocations and requests.
- b) Is the system in a deadlock state? Justify based on your RAG.
- c) If one instance of R2 is released by P4, describe the resulting changes in the graph and process states.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 3) A computer system employs paging for memory management, featuring a 30-bit logical address space and a page size of 16 KB. The physical memory available is 1 GB in size. (5)
- A) Determine how many bits in the logical address are allocated to the page number and how many to the page offset.
- Calculate the total number of page frames in the physical memory and find the number of bits required to represent the physical frame number.
- Lastly, given a specific page table mapping page numbers 0 to 3 to frame numbers 10, 15, 22, and 45 respectively, translate the logical address 0x00345FA2 for this process into its corresponding physical address.

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) Consider a computer system with the following characteristics: (5)
- Virtual address size: 40 bits
 - Page size: 16 KB
 - Each page table fits exactly in one page frame.
 - Page table size: 4 bytes
- Find:
- a) How many levels of page table will be required if each page table must fit into one page frame?
 - b) Calculate the physical memory size and calculate the total number of entries in the inverted page table. Determine the size of the inverted page table in bytes using above data.

Course Outcome: "CO5" / Cognitive Level: "Application"

- 4) **Given a system with 3-page frames and the following page reference string:** (5)
- A) **4, 1, 2, 4, 3, 2, 1, 5, 4, 2, 1, 6, 3**
- a) **Simulate the First-In-First-Out (FIFO) page replacement algorithm and determine the number of page faults. Draw the state of the page frames after each page reference.**
 - b) **Simulate the Least Recently Used (LRU) page replacement algorithm on the same reference string and calculate the number of page faults.**
 - c) **Compare the number of page faults from FIFO and LRU for this reference string and explain which algorithm performs better and why.**

Course Outcome: "CO5" / Cognitive Level: "Remember"

- B) Consider a system with the following parameters: (5)
- TLB access time = 20 ns
 - Main memory access time = 100 ns
 - The hit ratio of the TLB = 0.85
 - The page fault rate = 0.001
 - The page fault service time (including disk I/O) = 8 ms

Calculate the Effective Memory Access Time (EMAT)

- 5) A magnetic disk has several defining characteristics that influence its performance. The disk contains 2000 tracks, and the average seek time per track is 0.2 milliseconds. It spins at a rate of 3600 revolutions per minute (RPM). The data transfer rate is 200 megabytes per second, and the system transfers data in blocks of 4 kilobytes. Additionally, there is a controller overhead time of 8 milliseconds associated with disk operations. (5)
- A) Using these parameters, calculate the average seek time needed to position the read/write head, the average rotational latency representing the delay waiting for the disk to rotate to the correct position, the time taken to transfer the data block, and finally, the total average disk access time, to estimate overall disk access performance.

- B) A disk has 179 tracks. The disk arm is currently at track 64. The queue of requests in order is: (5)
89,129,61,139,51,19,158,12089,129,61,139,51,19,158,120
- Using the above, answer the following:
- a) Calculate the total head movement (seek time) and show the order of servicing requests using:
- C-SCAN Algorithm
 - LOOK Algorithm
 - C-LOOK Algorithm
 - SCAN Algorithm
 - SSTF Algorithm
- b) Which algorithm provides the best average seek time for this request set? Justify briefly.

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

ECM_3123:Operating Systems

#	Course Outcome	Description	Marks
1	2	CO 2	5
2	3	CO 3	5
3	4	CO 4	10
4	5	CO 5	20
5	6	CO 6	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

ECM_3123:Operating Systems

#	Level	No of Questions	Marks (%)
1	Remember	3	15(30%)
2	Understand	0	0(0%)
3	Application	3	15(30%)
4	Analysis	4	20(40%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)